

PIVOTVAULT

AI-Powered Startup Failure Intelligence Platform

PROJECT SPECIFICATION & ARCHITECTURAL REPORT

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EXECUTIVE SUMMARY

Over 90% of venture-backed startups fail, evaporating billions in capital due to repeating known failure modes (unit economics collapse, lack of PMF, premature scaling). PivotVault organizes 419+ verified failure autopsies, empowering founders with RAG-driven risk benchmarking before building.

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1. THE PROBLEM: THE BILLION-DOLLAR AMNESIA

Every year, thousands of founders embark on startups with sincere conviction, only to crash into the exact failure vectors that killed dozens of predecessors:

- Unverifiable Advice: Founders rely on generic survivorship-bias advice from successful founders whose playbook is no longer viable in the current macro climate.
- Scattered Evidence: Post-mortems, SEC bankruptcy filings, and customer complaints are fragmented across blogs, dockets, and forums with zero standardization.
- Premature Scaling: Teams expand burn and marketing before validating week-4 retention or positive unit economics, repeating the Kozmo, Fast, and Quibi mistakes.

2. THE SOLUTION: PIVOTVAULT FORENSIC INTELLIGENCE

PivotVault synthesizes historical failure data, machine learning risk classification, and Google Gemini retrieval-augmented generation to provide defensive founder intelligence:

- 419+ Verified Autopsies: Searchable database categorized across 15 industries and 14 lethal failure modes with capital, years, and root cause attribution.
- Idea Risk Scanner: Interactive radar tool computing an objective Risk Score (0-100) and surfacing the top 3 nearest historical failures via vector cosine distance.
- Hall of Ghosts: Conversational AI reconstructs founder personas from public testimony so founders can interrogate past decision points directly.
- Diagnostic Autopsy Suite: Pitch deck audit, side-by-side competitor comparison, and defensive counter-moves formulated from historical precedent.

3. TECHNICAL ARCHITECTURE & STACK

- Frontend: Vite 6, React 19, Tailwind CSS, Framer Motion, Lucide Icons.
Fast static distribution on Vercel with offline curated datasets.
- Backend API: Node.js 20, Express, Prisma ORM, BullMQ distributed queue workers.
- Database: PostgreSQL 16 with pgvector extension (768-dimension Gemini embeddings).
- ML Service: Python FastAPI, Scikit-learn IdeaScore classifier, CatBoost failure weighting.
- Large Language Models: Google Gemini 1.5 Flash (RAG reasoning) & Groq LLaMA 3.3 70B.

4. VERIFICATION & RESULTS

- 100% Production Build Pass Rate on Vite 6.
- Sub-10ms similarity queries across 419+ indexed vectors using IVFFlat indices.
- Graceful 3-second offline fallback ensuring 100% uptime for platform reviewers.
- Cross-referenced against canonical venture datasets: CB Insights, Failory, PitchBook.

Developed by Team CodeRegime (KH024) for the Startup Innovation Challenge.